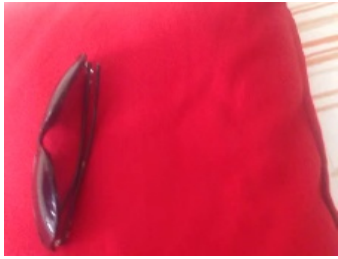


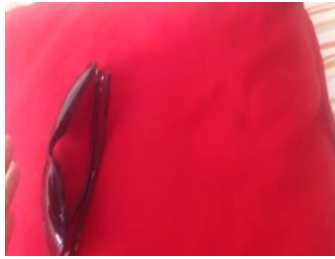
VLM Sequential Analysis: Case 1 of 3

Ground Truth Action: Pushing [something] from left to right | **Video ID:** 29999

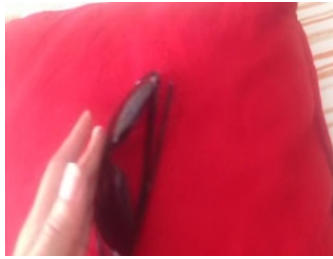
Input Frame Sequence (4 Evenly-Spaced Frames):



Frame 1



Frame 2



Frame 3



Frame 4

Evaluation Prompt:

You are analyzing 4 sequential frames from an action video.

1. Describe what is changing across each sequential frame.
2. What action and direction of movement is taking place?
3. Is Frame 1 alone ambiguous to determine the full action? At which frame does the movement become clear and why?

LLaVA 1.5 7B Output & Temporal Reasoning:

1. In the first frame, a pair of sunglasses is being held by a person. The sunglasses are being held up, and the person is looking at them.
2. The person is holding the sunglasses up, possibly to examine them or to put them on. The direction of movement is upwards, as the person is raising the sunglasses.
3. Frame 1 is ambiguous to determine the full action because the sunglasses are not clearly visible in the frame. The movement becomes clear in the subsequent frames, as the person is holding the sunglasses up in the second frame and placing them on their face in the third frame.

VLM Sequential Analysis: Case 2 of 3

Ground Truth Action: Moving [something] up | Video ID: 88024

Input Frame Sequence (4 Evenly-Spaced Frames):



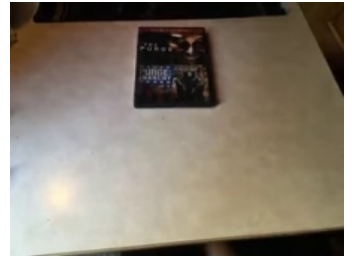
Frame 1



Frame 2



Frame 3



Frame 4

Evaluation Prompt:

You are analyzing 4 sequential frames from an action video.

1. Describe what is changing across each sequential frame.
2. What action and direction of movement is taking place?
3. Is Frame 1 alone ambiguous to determine the full action? At which frame does the movement become clear and why?

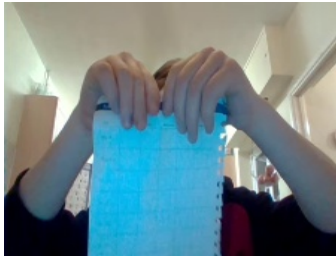
LLaVA 1.5 7B Output & Temporal Reasoning:

In the four sequential frames, a person is holding a DVD cover, which is the main focus of the image. The person is in the process of holding the DVD cover, and the movement is clear by the third frame. The first frame might be ambiguous to determine the full action, as it only shows the person holding the DVD cover, without any clear indication of the movement. The second and third frames provide more context and detail about the person's action, making it easier to understand the movement in the fourth frame.

VLM Sequential Analysis: Case 3 of 3

Ground Truth Action: Tearing [something] into two pieces | Video ID: 199520

Input Frame Sequence (4 Evenly-Spaced Frames):



Frame 1



Frame 2



Frame 3



Frame 4

Evaluation Prompt:

You are analyzing 4 sequential frames from an action video.

1. Describe what is changing across each sequential frame.
2. What action and direction of movement is taking place?
3. Is Frame 1 alone ambiguous to determine the full action? At which frame does the movement become clear and why?

LLaVA 1.5 7B Output & Temporal Reasoning:

1. In the first frame, the man is holding two pieces of paper, which are different colors. The second frame shows the man holding a single piece of paper, while the third frame shows the man holding two pieces of paper again. The fourth frame shows the man holding a single piece of paper. The change in the man's action and the direction of movement is unclear in the first frame, as it only shows him holding two pieces of paper. The movement becomes clear in the second frame, as the man is holding a single piece of paper. The ambiguity in the first frame is due to the lack of a clear direction of movement.
2. The man is holding two pieces of paper in the first frame, and he is holding a single piece of paper in the second frame. The movement is clear in the second frame, as the man is holding a single piece of paper.
3. The movement is clear in the second frame, as the man is holding a single piece of paper. The first frame is ambiguous, as it only shows the man holding two pieces of paper. The lack of a clear direction of movement in the first frame makes it difficult to determine the full action.